Detecting Colon and Lung Cancer through Deep Learning CNN Model

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Abstract— **Lung and colon cancers pose substantial global health challenges, being recognised as prevalent and potentially life-threatening tumours. This research focuses on the development and evaluation of a DL model that utilises Convolutional Neural Networks to classify images of lung and colon cancer into five distinct categories. By making use of a dataset consisting of 25,000 medical images, the CNN model was subjected to a rigorous 40-epoch training process with a batch size of 40. The noteworthy outcome of this investigation is the remarkable achievement of a 99% accuracy rate by the CNN model, showcasing the transformative potential of DL in medical image analysis. The implications of such high accuracy extend to early diagnosis, therapy planning, and overall patient care in the medical field. This paper underscores the versatility of CNN architecture in healthcare applications, emphasising its significance in addressing complex medical classification tasks. The importance of automation and accuracy in clinical settings is highlighted, urging further research and validation of deep learning methodologies. The study demonstrates the effectiveness of CNN models in cancer classification, thereby facilitating the advancement of medical image processing systems that are more precise and effective. The end result will be advantageous for healthcare practitioners and patients equally.**

*Keywords— **Deep learning, Convolutional Neural Network, Lung cancer, Colon cancer, Cancer diagnosis, Medical imaging, Early detection.***

I. INTRODUCTION

On a global scale, lung and colon cancer are globally recognised as highly prevalent and potentially fatal diseases, posing significant challenges to the domain of public health. Timely identification of malignant neoplasms is of the utmost importance in maximising survival rates and enhancing prognoses for those affected. The respiratory system is the primary target of lung cancer, while the large intestine is the organ of genesis for colon cancer. Both illnesses have the potential to remain asymptomatic in their first stages, presenting a considerable obstacle in the prompt detection and identification of these ailments. Lung cancer is a substantial and possibly lethal disease that is well-recognised as a major contributor to worldwide cancer-related deaths. Lung cancer is a pathological state characterised by the abnormal and uncontrolled growth of atypical cells in the pulmonary tissues, often leading to the development of tumours that can impede essential respiratory functions. The formidable challenge posed by lung cancer is attributed to its aggressive behaviour, propensity for metastasis, and often delayed detection. Acquiring a thorough comprehension of the essential elements of this ailment is imperative in the ongoing efforts to improve the effectiveness of preventive measures, facilitate timely detection, and optimise treatment results. Lung cancer is classified into two primary subtypes, namely small cell lung cancer and non-small cell lung cancer, each possessing distinct attributes and necessitating tailored approaches to therapy. The manifestation of symptoms associated with lung cancer may exhibit intricacy in the early stages, often leading to a delay in detection. However, as the disease progresses, the ensuing manifestations and symptoms may become apparent. The patient exhibits several symptoms, such as a chronic cough, dyspnea, thoracic discomfort, unintentional weight loss, exhaustion, dysphonia, recurrent respiratory infections, skeletal pain, and cephalalgia. Lung cancer incidence is widely recognised to be correlated with tobacco use, with smoking being identified as the primary causative factor in a substantial proportion of detected cases. The inhalation of smoke leads to the exposure of the lungs to detrimental substances, hence inducing genetic alterations that trigger the uncontrolled proliferation of cells, a characteristic hallmark of carcinogenesis. However, those who choose to remain tobacco-free may remain vulnerable to the onset of lung cancer, a disease frequently linked to a variety of underlying factors. The opportune detection of lung cancer substantially improves the prognosis. Various methodologies are employed to ascertain the presence of a medical condition. Imaging modalities for diagnostic purposes: CT scans and chest X-rays are utilised in order to detect abnormalities in the lungs. Low-dose CT scans are particularly advantageous for the early detection of health issues in persons at a heightened risk, such as those who engage in continuous smoking. Sputum cytology involves the examination of expectorated lung material, commonly referred to as sputum. The specimen is subjected to microscopic examination in order to identify any aberrant cellular structures. Biopsy: A tissue sample is obtained through medical procedures like surgery, needle biopsy, or bronchoscopy in order to determine the type and presence of cancer. Analysing tumour samples genetically to identify specific genetic abnormalities that may influence the choice of targeted therapeutic interventions is known as molecular

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testing. Colon cancer, also known as colorectal cancer, exerts its impact on the colon and rectum within the digestive system. The aforementioned condition is widely prevalent and has the potential to result in mortality. It ranks as the third most frequently diagnosed cancer globally in both males and females. Failure to undergo treatment for colon cancer can result in the progression of benign polyps, which are tiny growths located in the colon or rectum lining, into malignant tumours. Due to the elevated prevalence of this particular form of cancer, its detrimental effects on individuals' quality of life, and its capacity to metastasise to many organs, it presents a significant public health concern. Colon cancer typically exhibits a slow onset over several years, providing opportunities for early detection and intervention. Modifications in bowel habits, instances of rectal haemorrhaging, indications of abdominal distress or pain, unexplained weight loss, and feelings of fatigue are frequently documented symptoms. Noting that the manifestation of symptoms may be negligible or absent in particular instances is crucial. In cases where these symptoms manifest with significant intensity, a medical evaluation becomes imperative. The primary determinant of susceptibility to colon cancer is age, although there exist additional contributing factors. The majority of cases are diagnosed in individuals aged 50 years and beyond. Additional risk factors for colorectal cancer include a diet characterised by a high intake of processed meats and a low intake of fibre, being overweight or obese, Individuals who smoke cigarettes or consume alcohol in excessive quantities, have a family history of colorectal cancer or tumours, and suffer from underlying conditions such as inflammatory bowel disease (IBD), are at increased risk. The successful management of colon cancer is contingent upon the timely detection of the disease. Colonoscopies, sigmoidoscopies, faecal occult blood tests, and virtual colonoscopies are often employed screening modalities for the detection and removal of polyps or the identification of cancer at an earlier and more amenable stage. Considering the individual's age and risk factors, the optimal frequency of these examinations will vary. The type of treatment for colon cancer varies on a number of things, such as where the tumour is located in the body, how big it is, what stage it is in, and the patient's general health. Surgery is commonly employed as the primary therapeutic intervention, wherein the tumour is excised together with adjacent tissue. Chemotherapy, radiation therapy, targeted therapies, and immunotherapy represent a range of therapeutic interventions employed to diminish tumour dimensions, eradicate malignant cells, or impede their proliferation. Lung and colon cancers are widely observed and extremely aggressive tumours that have a substantial worldwide influence. The prompt detection of cancer is crucial for improving patient outcomes, as the efficacy of treatment is significantly impacted by the timing of tumour diagnosis. This work presents a unique CNN architecture designed to facilitate the efficient categorisation of lung and colon cancer by analysing the medical imaging data. Examining and classifying radiographic images, including X-rays and CT scans, in an effort to detect malignant abnormalities, the proposed CNN model employs deep learning techniques. The current study employs a comprehensive compilation of medical pictures that have been carefully chosen to include a diverse range of patients, including various stages of L&C cancer. In order to classify input images with a substantial degree of precision, the CNN architecture was purposefully designed to extract prominent features. The efficiency of the model is assessed by rigorous validation methods, including cross-validation and external testing, to determine its capacity for generalisation and stability.

This article makes several significant contributions. The fundamental objective of this work is to construct and assess a novel CNN structure that is specifically engineered to precisely detect and classify lung (L) and colon (C) cancers through the utilisation of medical imaging data. Colonic cancer, benign tissue of the colon, benign tissue of the lung, adenocarcinoma of the lung, and squamous cell carcinoma of the lung are the five distinct categories that the dataset has carefully categorised. The proposed CNN structure undergoes 40 iterations of training, each with a batch size of 40. This configuration demonstrates a meticulously planned and extensively optimised training schedule. This approach ensures the model's efficacy in learning and adapting to the intricate patterns within the medical imaging data. The evaluation of CNN's performance is carried out through the utilisation of Confusion Matrix performance metrics. This method provides a comprehensive and detailed assessment of the model's ability to correctly classify instances within each category, thus highlighting the robustness and reliability of the proposed architecture.

This comprehensive resource offers extensive treatment of all key themes. Section II of this paper shows a comprehensive literature analysis on the identification of datasets related to lung and colon cancer. Section III presents a comprehensive overview of the approaches that are recommended for use. The assemblage of findings and examination of data can be located in Section IV. The final paragraph of Section V provides clarification on the conclusion.

II. LITERATURE REVIEW

Mangal et al. (2020) [12] A study was done with the objective of creating a computerised diagnostic system that utilises CNN for the histological diagnosis of lung and colon tumours. The focus of the study was on squamous cell carcinomas and cancers of the in the lung, as well as cancers of the in the colon. Using a dataset of 2500 digital images, a shallow neural network architecture achieved diagnostic accuracies of more than 97% for lung cancer and 96% for colon cancer. The research demonstrates the potential of artificial intelligence in enhancing cancer diagnosis through digital pathology images, offering a promising avenue for future applications.

Masud et al. (2021) [13] The study utilised Artificial Intelligence (AI) in automating cancer diagnosis. It proposes a classification framework using Deep Learning and Digital Image Processing techniques to identify lung and colon cancers. The results show a 96.33% accuracy rate, demonstrating the potential of AI in aiding medical professionals in identifying these types of cancers.

Attallah et al. (2022) [14] This study proposes a novel framework utilising multiple lightweight DL models, including MobileNet, SqueezeNet and ShuffleNet, for early cancer detection. Transformation methods such as Discrete wavelet transform (DWT), principal component analysis, and fast Walsh–Hadamard transform are utilised to decrease the dimensionality of features and improve the representation of data. The reduced features from these methods are fused and fed into machine learning algorithms, achieving an impressive accuracy of 99.6%.

Tasnim et al. (2021) [15] The study employs CNN architecture that incorporates both max pooling and average pooling layers. Additionally, the MobileNetV2 model is employed for the purpose of classifying colon cell images. Training and evaluating these models at various Epochs reveal impressive accuracies, a maximum pooling accuracy of 97.49% and an average pooling accuracy of 95.48% are obtained, respectively, when these models are trained and evaluated at different epochs. It is worth mentioning that MobileNetV2 maintains a minimal data loss rate of 1.24% and achieves an exceptional accuracy rate of 99.67%. This work highlights the potential of CNN models, particularly MobileNetV2, in automating and enhancing the accuracy of colon cancer diagnoses.

Wahid et al. (2023) [16] This research paper introduces a computer-aided diagnostic system that utilises CNN for the detection of lung and colon cancer cells. The system is evaluated using the LC25000 dataset, which comprises 25,000 histopathological colour picture samples. CNN models that have been pre-trained include GoogLeNet, ResNet18, and ShuffleNet V2, along with a customised CNN model employed. ResNet18 demonstrated the highest accuracy for lung cancer at 98.82%, while ShuffleNet V2 excelled in colon cancer classification with a 99.87% accuracy. The customised CNN model demonstrated an accuracy of 88.26% for colon cancer and 93.02% for lung cancer, along with the shortest training time among the models, outperforming GoogLeNet and ResNet18. This study shows that CNN models are very good at automatically finding lung and colon cancers, providing valuable insights into accuracy and training efficiency.

Hadiyoso et al. (2023) [17] In this study, A CNN with Contrast Limited Adaptive Histogram Equalisation (CLAHE) and VGG16 architecture is employed to develop a DL-based automatic detection method for colon and lung cancer. The classification involves normal tissue, benign cancer, and malignant cancer histopathological images. As indicated by the utmost accuracy of 98.96% in the simulation results, the proposed procedure is effective.

Saleh et al. (2021) [18] This work introduces a hybrid classification method employing a combination of Support Vector Machine and CNN to automatically analyse lung CT images for signs of cancer. The CNN, known for efficient training and parameter reduction, is complemented by SVM to eliminate irrelevant information and enhance accuracy. The proposed CNN-SVM algorithm demonstrates exceptional performance, achieving a classification accuracy of 97.91% in evaluating lung images. The study underscores the efficacy of the hybrid methodology in effectively identifying potential cancer cells, emphasising its potential as a robust Computer-Aided Detection (CAD) tool for lung cancer diagnosis.

III. METHODOLOGY

Figure 1 represents the architectural configuration of the CNN model. The dataset encompasses 25,000 photos meticulously sorted into five main categories, each comprising 5,000 images. The data-gathering process utilised the CNN Deep Learning model with rescaling techniques. The dataset utilized in this research was acquired through the CNN model specified on the Kaggle platform. A comprehensive explanation of each category follows.

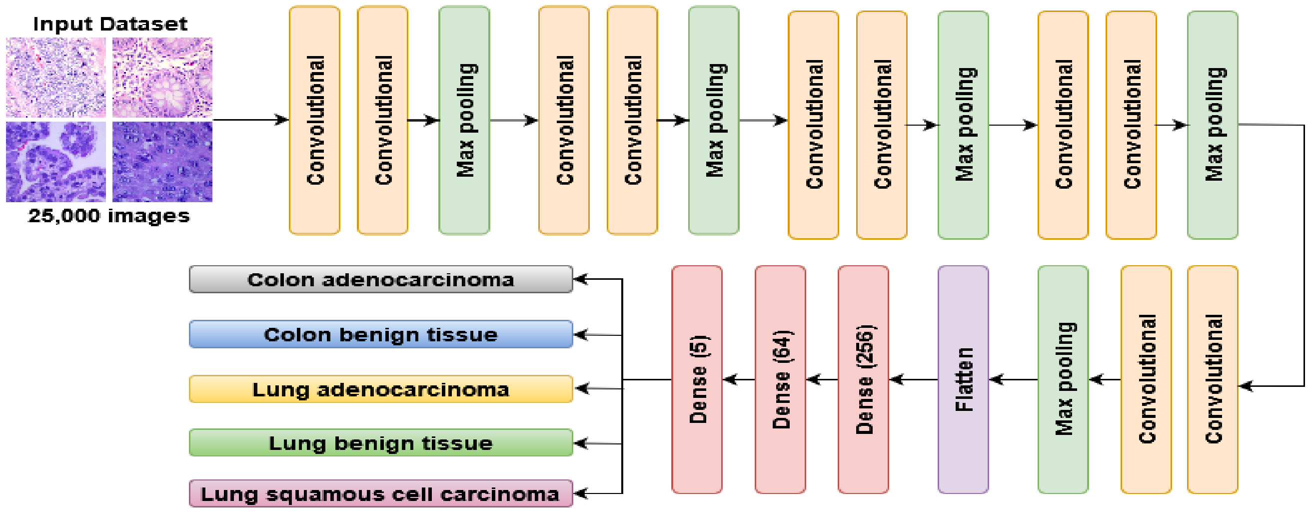

Figure 1 Proposed Methodology

A. Dataset Description

The dataset for corn or maize plant leaf disease classification consists of 4388 images, with each class representing specific disease conditions: Common Rust (1306 images), Gray Leaf Spot (574 images), Blight (1146 images), and Healthy leaves (1162 images). This balanced distribution mitigates class imbalance issues. The dataset is split into training (3333 images) and testing (1429 images) sets for model training and evaluation. Fig 2 illustrates sample images from the dataset, providing a visual overview of the diverse leaf conditions. This curated dataset serves as the foundation for subsequent random oversampling and the implementation of the EfficientNet-B1 model, ensuring accurate and robust disease classification.

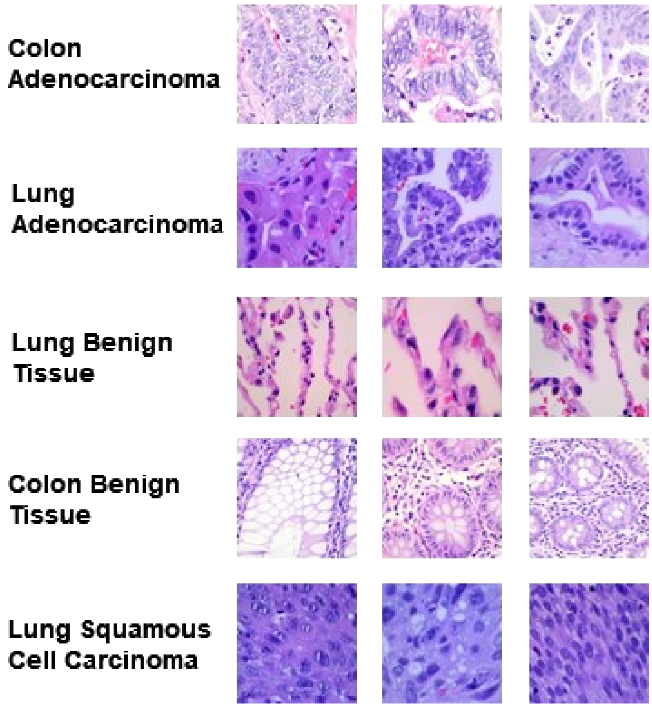

Figure 2 Input Dataset

B. Data Augmentation and Processing

The dataset used for classifying maize leaf diseases is subjected to thorough data processing and augmentation techniques in order to enhance its compatibility with deep learning algorithms. The initial procedures involve the scaling of all photos to a uniform dimension of 224x224 pixels and the conversion of colour schemes from BGR to RGB. In order to mitigate the issue of class imbalance, a random oversampling strategy is employed, which specifically targets the minority class known as Grey Leaf Spot. This technique involves replicating and enhancing pictures belonging to the minority class. The process of shuffling data during model training adds a level of unpredictability that serves to impede the learning of sequential patterns. The process of normalising pixel values to the range of [0, 1] has been shown to improve the convergence rate during training. The implementation of augmentation techniques, such as rotation and flips, serves to enhance the variety of the dataset. The aforementioned steps combined yield a dataset that is balanced, diversified, and resilient, hence establishing the necessary groundwork for the successful training and assessment of the maize leaf disease classification model.

C. Model Architecture

The model architecture, as seen in Figure 3, is based on the EfficientNet-B1 convolutional neural network (CNN), which has been pre-trained on the ImageNet dataset. The architectural design of this system incorporates adaptive scaling and compound scaling approaches, which enhance its capacity to effectively capture complex patterns within visual data. The fundamental architecture includes convolutional layers, global average pooling, dropout regularisation, and densely linked layers with rectified linear unit (ReLU) activation. The ultimate dense layer, which utilises softmax activation, produces predictions of class probabilities. It is worth mentioning that, in order to prevent overfitting, the layers of the basic model are kept non-trainable during the process of fine-tuning. The strategic option made in this context effectively achieves a delicate balance between the complexity of the model and its efficiency. The model achieves precise and resilient predictions by effectively incorporating pre-existing knowledge with the distinctive attributes of maize leaf diseases.

D. *Proposed Work*

This study presents a unique methodology that combines random oversampling with the transfer learning capabilities of the EfficientNet-B1 architecture to improve the classification of maize leaf diseases. The methodology being presented leverages the efficacy of oversampling techniques, notably random oversampling, in order to effectively mitigate the issue of class imbalance present in the dataset. The strategy seeks to promote a fairer representation of each class by selectively increasing the size of the minority class. This is done with the intention of reducing biases in the learning process of the model.

In addition, the use of the advanced EfficientNet-B1 transfer learning model in illness classification aims to enhance the model's ability to accurately distinguish between different diseases. EfficientNet-B1 leverages adaptive scaling and compound scaling techniques to enhance its ability to capture intricate characteristics present in maize leaf diseases, hence facilitating the generation of reliable and precise predictions. The utilisation of random oversampling in conjunction with EfficientNet-B1 serves to address the issue of class imbalance while also enhancing the model's comprehension of the complex patterns present in the dataset. By means of this synergistic fusion, the present study aims to enhance the effectiveness of maize leaf disease categorization, so offering a valuable contribution to the generation of more dependable and comprehensible outcomes in the context of agricultural decision-making.

C. Accuracy and Loss Plots

This section conducts a comprehensive analysis of visual representations portraying the Accuracy and Loss plots, utilising the CNN model as the chosen methodology. Figure 3 visually encapsulates this methodology, with the red line signifying training loss and accuracy and the green line depicting validation loss and accuracy. Additionally, the blue dot on the graph identifies the epoch with the highest performance. Figure 3(a) offers a detailed view of accuracy in both the training and validation phases, while Figure 3(b) illustrates the corresponding loss across the training and validation processes. The visual representations proficiently communicate the correlation between the number of training epochs and model accuracy, thereby offering a useful tool for visually illustrating the compromise between precision and generalisability.

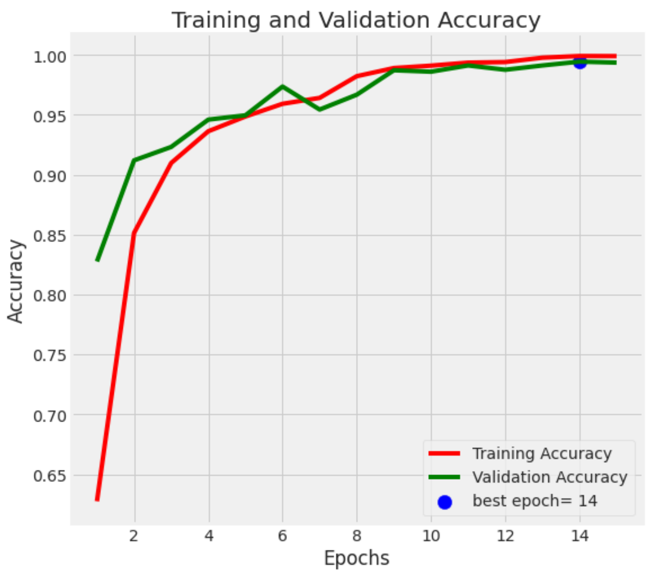

Figure 3(a) Accuracy Plots

Figure 3(b) Loss Plots

Figure 3. Training and Test Curve (a)Accuracy and (b)Loss

D. Confusion Matrix

Figure 4 provides a graphical representation of the confusion matrix's parameters. The creation of a 5x5 matrix correlates to the various attributes associated with five specific kinds of lung and colon cancers. The utilisation of a confusion matrix is a valuable approach for visually representing the accuracy of classification. The evaluation of a system's performance can be exemplified by employing performance metrics such as accuracy, recall, precision, and F1-score. The results of the evaluation of performance are presented in Table 1.

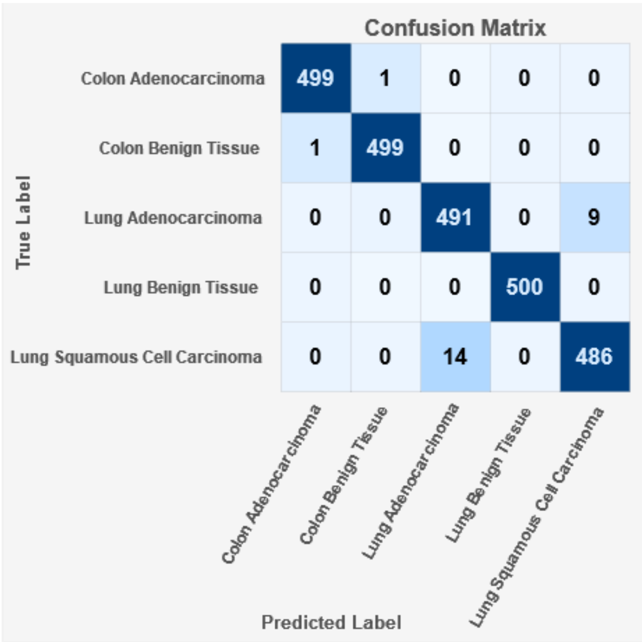

Figure 4 Confusion Matrix

E. Performance Parameter

The classification of the images is simplified through the application of the confusion matrix, which is illustrated in Table 1. Lung and colon cancer are represented by five distinct categories in the matrix. The assessment of the CNN model's effectiveness in classification requires the application of the Confusion Matrix. The matrix illustrated above serves as a mechanism for evaluating the network's efficacy in precisely differentiating among five discrete subtypes of colon and lung cancer. The assessment of the CNN model will involve the analysis of many metrics, such as accuracy, precision, F1 score, and recall, among other factors. The data points were compared in order to assess their level of accuracy, which was found to reach up to 99%.

Table 1 Performance Parameter

Name of the Classes	Precision	Recall	F1-Score	Accuracy
Colon Adenocarcinoma	100%	100%	100%	
Colon Benign Tissue	100%	100%	100%	
Lung Adenocarcinoma	97%	98%	98%	99%
Lung Benign Tissue	100%	100%	100%	
Lung Squamous Cell Carcinoma	98%	97%	98%	

IV. CONCLUSION

This work introduces a CNN model that utilises deep learning techniques to accurately identify datasets associated with colon and lung cancer. The CNN model has proven to be highly effective in this regard. The training procedure consisted of 40 epochs, with a batch size of 40 for each epoch. During the training process, the Adam and Adamax optimisers were utilised. Achieving an accuracy rate of 99% is a notable accomplishment. The observed high level of precision exhibited by CNN highlights their potential to advance the fields of medical image processing and cancer diagnosis, which could have significant ramifications for the medical community. The utilisation of deep learning methodologies to precisely categorise various forms of cancer shows the capacity to enhance early detection, advance the diagnosis process, and potentially optimise treatment outcomes. While the results obtained demonstrate potential, it is imperative to emphasise the necessity for further testing, clinical studies, and integration into existing systems to provide comprehensive validation. The work presented in this study highlights the significant potential of DL models in the field of healthcare, particularly in improving healthcare outcomes and mitigating the global burden of cancer. The model provided in this study exemplifies the ongoing progress and refinement of DL techniques. The primary objective of this endeavour is to foster further progress at the convergence of the medical and artificial intelligence fields.

REFERENCES

[1] A. M. Godkhindi and R. M. Gowda, “Automated detection of polyps in CT colonography images using deep learning algorithms in colon cancer diagnosis,” in 2017 International Conference on Energy, Communication, Data Analytics and Soft Computing (ICECDS), 2017.

[2] X. Zhang, K. P. Rane, I. Kakaravada, and M. Shabaz, “Research on vibration monitoring and fault diagnosis of rotating machinery based on internet of things technology,” Nonlinear Eng., vol. 10, no. 1, pp. 245–254, 2021.

[3] S. Bharany, “Energy efficient fault tolerance techniques in green cloud computing: A systematic survey and taxonomy, ” Sustain,” Sustain. Energy Technol. Assessments, vol. 53, no. 102613, 2022.

[4] M. Yang, P. Kumar, J. Bhola, and M. Shabaz, “Development of image recognition software based on artificial intelligence algorithm for the efficient sorting of apple fruit,” Int. J. Syst. Assur. Eng. Manag., vol. 13, no. S1, pp. 322–330, 2022.

[5] Sakshi and V. Kukreja, “A retrospective study on handwritten mathematical symbols and expressions: Classification and recognition,” Eng. Appl. Artif. Intell., vol. 103, no. 104292, p. 104292, 2021.

[6] V. Anand, S. Gupta, D. Koundal, S. R. Nayak, P. Barsocchi, and A. K. Bhoi, “Modified U-NET architecture for segmentation of skin lesion,” Sensors (Basel), vol. 22, no. 3, p. 867, 2022.

[7] J. Singh, D. Thakur, F. Ali, T. Gera, and K. S. Kwak, “Deep feature extraction and classification of Android Malware images,” Sensors (Basel), vol. 20, no. 24, p. 7013, 2020.

[8] V. Anand, S. Gupta, D. Koundal, and K. Singh, “Fusion of U-Net and CNN model for segmentation and classification of skin lesion from dermoscopy images,” Expert Syst. Appl., vol. 213, no. 119230, p. 119230, 2023.

[9] M. Pundir and J. K. Sandhu, “A systematic review of quality of service in wireless sensor networks using machine learning: Recent trend and future vision,” J. Netw. Comput. Appl., vol. 188, no. 103084, p. 103084, 2021.

[10] S. M. Sasubilli, A. Kumar, and V. Dutt, “Machine learning implementation on medical domain to identify disease insights using TMS,” in 2020 International Conference on Advances in Computing and Communication Engineering (ICACCE), 2020.

[11] A. Kumar, J. M. Chatterjee, and V. G. Díaz, “A novel hybrid approach of SVM combined with NLP and probabilistic neural network for email phishing,” Int. J. Electr. Comput. Eng. (IJECE), vol. 10, no. 1, p. 486, 2020.

[12] S. Mangal, A. Chaurasia, and A. Khajanchi, “Convolution Neural Networks for diagnosing colon and lung cancer histopathological images,” arXiv [eess.IV], 2020.

[13] M. Masud, N. Sikder, A.-A. Nahid, A. K. Bairagi, and M. A. AlZain, “A machine learning approach to diagnosing lung and colon cancer using a Deep Learning-based classification framework,” Sensors (Basel), vol. 21, no. 3, p. 748, 2021.

[14] O. Attallah, M. F. Aslan, and K. Sabanci, “A framework for lung and colon cancer diagnosis via lightweight deep learning models and transformation methods,” Diagnostics (Basel), vol. 12, no. 12, p. 2926, 2022.

[15] Z. Tasnim et al., “Deep learning predictive model for colon cancer patient using CNN-based classification,” Int. J. Adv. Comput. Sci. Appl., vol. 12, no. 8, 2021.

[16] R. R. Wahid, C. Nisa, R. P. Amaliyah, and E. Y. Puspaningrum, “Lung and colon cancer detection with convolutional neural networks on histopathological images,” in AIP Conference Proceedings, vol. 2654, AIP Publishing, 2023.

[17] S. Hadiyoso, S. Aulia, and I. D. Irawati, “Diagnosis of lung and colon cancer based on clinical pathology images using convolutional neural network and CLAHE framework,” International Journal of Applied Science and Engineering, vol. 20, no. 1, pp. 1–7, 2023.

[18] A. Y. Saleh, C. K. Chin, V. Penshie, and H. R. H. Al-Absi, “Lung cancer medical images classification using hybrid CNN-SVM,” Int. J. Adv. Intell. Inform., vol. 7, no. 2, p. 151, 2021.